QING DAI

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Education

Ph.D. Candidate in Bioengineering 2020 – Present
University of California, Los Angeles
Topics: *MRI-guided Interventions*
Advisor: Dr. Holden H. Wu

M.S. in Biomedical Imaging 2018 – 2019
University of California, San Francisco
Topic: *Deep Learning-Based Tumor Grade Prediction*
Advisor: Dr. Peder E. Z. Larson

B.S. in Biochemistry, Minor in Bioinformatics 2013 – 2017
University of California, Los Angeles

Professional Experience

University of California, Los Angeles 2020 – Present
Graduate Student Researcher
- Developing real-time MRI and MR thermometry techniques for MRI-guided minimally invasive interventions.
- Developing MR-guided nanoparticle drug delivery platforms and cancer theranostics.
- Leading experimental design, execution, and advanced data analysis across benchtop phantom, pre-clinical animal models (mouse and pig), and human subjects.
- Collaborating with a multi-disciplinary team of radiologists, physicians, engineers, chemists, veterinarians, and industrial collaborators toward clinical translation.

University of California, Los Angeles 2025
Technology Fellow, UCLA Technology Development Group
- Drafting technology non-confidential disclosures and identifying marketing targets to facilitate university technology licensing and commercialization.

University of California, San Francisco 2018 – 2019
Research Specialist
- Developed quantitative models for renal tumor cells detection, classification, and aggressiveness prediction from contrast-enhanced CT and clinical metadata.
- Built kinetic models and metabolic quantification methods for dynamic hyperpolarized carbon-13 pyruvate MRI.

University of California, Los Angeles 2016 – 2018
Research Assistant
- Developed a high-throughput image processing and cell counting pipeline for immunohistochemistry using machine learning techniques.

Research Interests

Image-guided Therapy	MR Thermometry	Thermal Ablation
Computational Modeling	Image Reconstruction	Nano Theranostics

Academic Societies

International Society in Magnetic Resonance in Medicine **(ISMRM)**

Institute of Electrical and Electronics Engineers**(IEEE)**

Focused Ultrasound Foundation **(FUS)**

International Society for Therapeutic Ultrasound **(ISTU)**

Honors

Bioengineering Graduate Fellowship, UCLA Graduate Division	2026
Magna Cum Laude Merit Award, ISMRM	2026
Trainee Travel Award, Jonsson Comprehensive Cancer Center	2025
AMPC Selected Award, ISMRM	2024–2025
Summa Cum Laude Merit Award, ISMRM	2023–2025
Educational Stipend, ISMRM	2023–2025
Interventional MR Study Group Award (2nd Place), ISMRM	2023
Bioengineering Department Fellowship, UCLA Graduate Division	2020
Outstanding Poster Award, Hyperpolarized Carbon-13 MRI Workshop	2020

Professional Service

Reviewer and Code Reviewer, Magnetic Resonance in Medicine	2024 - 2026
Abstracts Reviewer, ISMRM Annual Meeting	2023 - 2026
Organizer, MRRL Journal Club	2022 - 2026

Journal Publications

1. **Dai Q**, Chiang J, Shih SF, Zhou W, Lu DSK, Wu HH. Active electromagnetic interference suppression for MR thermometry during MR-guided microwave ablation. *Magnetic Resonance in Medicine*. 2026. doi: 10.1002/mrm.70440
2. Sahin S, Diaz E, Rajagopal A, Abtahi M, Jones S, **Dai Q**, Kramer S, Wang Z, Larson PEZ. UCSF RMaC: A University of California San Francisco 3D Multi-Phase Renal Mass CT Dataset with Tumor Segmentations. Preprint available at *medRxiv*. doi:10.1101/2026.02.11.26346096.
3. Zhou W, **Dai Q**, Curiel O, Tsao TC, Chiang J, Lu DS, Wu HH. Keypoint detection network for needle localization on intra-procedural MRI in MRI-guided liver interventions. *International Journal of Computer Assisted Radiology and Surgery*. 2026. doi:10.1007/s11548-026-03592-5.
4. **Dai Q**, Shih SF, Curiel O, Tsao TC, Lu DS, Chiang J, Wu HH. Volumetric thermometry in moving tissues using stack-of-radial MRI and an image-navigated multi-baseline proton resonance frequency shift method. *Magnetic Resonance in Medicine*. 2026;95(2):803-819. doi:10.1002/mrm.70074.
5. Deng T†, **Dai Q**†, Estabrook DA, Chapman J, Sletten EM, Wu HH, Zink JI. Thermoresponsive polymer-modified SiO_2 / gadolinium-diethylenetriamine pentaacetic acid composite nanoparticles for magnetic resonance imaging-guided ultrasound-modulated contrast enhancement at human body temperatures. *ACS Applied Nano Materials*. 2025;8(47):22835-22844. doi:10.1021/acsanm.5c04320 († equal contribution).
6. Sahin S, Haller AB, Gordon J, Kim Y, Hu J, Nickles T, **Dai Q**, Leynes AP, Vigneron DB, Wang ZJ, Larson PEZ. Spatially constrained hyperpolarized ^{13}C MRI pharmacokinetic rate constant map estimation using a digital brain phantom and a U-Net. *Journal of Magnetic Resonance*. 2025;371:107832. doi:10.1016/j.jmr.2025.107832.
7. Zhong X, Nickel MD, Kannengiesser SAR, Dale BM, Han F, Gao C, Shih SF, **Dai Q**, Curiel O, Tsao TC, Wu HH, Deshpande V. Accelerated free-breathing liver fat and R_2^* quantification using multi-echo stack-of-radial MRI with motion-resolved multidimensional regularized reconstruction: Initial retrospective evaluation. *Magnetic Resonance in Medicine*. 2024;92(3):1149-1161. doi:10.1002/mrm.30117.1.

Conference Papers

8. Li J, Hou Q, Pang K, Miao Q, Hung ALY, Aygun E, Shih SF, **Dai Q**, Wu HH, Sung KH. CoDe: A Self-Supervised Consistency Model Framework for MRI Denoising. *2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*, 2026. doi: 10.1109/ISBI61048.2026.11515485.
9. Pham B, Gaonkar B, Whitehead W, Moran S, **Dai Q**, Macyszyn L, Edgerton VR. Cell counting and segmentation of immunohistochemical images in the spinal cord: Comparing deep learning and traditional approaches. *2018 40th Annual International*

Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2018. pp. 842-845. doi: 10.1109/EMBC.2018.8512442.

Patents

1. Wu HH, **Dai Q**. Active Electromagnetic Interference Suppression for Magnetic Resonance Imaging-Guided Intervention. U.S. Provisional Patent Application, filed March 2026. (Patent Pending)
2. Wu HH, Zhou W, **Dai Q**, Lu DS. Keypoint Detection Network for Device Localization on Intra-procedural MRI. U.S. Provisional Patent Application, filed March 2026. (Patent Pending)

Conference Abstracts

1. **Dai Q**, Chiang J, Zhou W, Lu D, Wu HH. Characterizing the spatiotemporal thermal effects of pulsed microwave ablation with computational modeling and MR thermometry. *International Society for Magnetic Resonance in Medicine Annual Meeting, 2026, Cape Town, South Africa.* **(Magna Cum Laude)**
2. Zhou W, **Dai Q**, Kerr CR, Shih SF, Delgado T, Tsao TC, Lu DS, Chiang J, Wu HH. End-to-end framework for real-time image reconstruction, device and tissue tracking in MRI-guided liver interventions. *International Society for Magnetic Resonance in Medicine Annual Meeting, 2026, Cape Town, South Africa.* **(Magna Cum Laude)**
3. **Dai Q**, Shih SF, Chiang J, Lu DS, Wu HH. Active electromagnetic interference suppression for real-time MR thermometry during MR-guided microwave ablation. *International Society for Magnetic Resonance in Medicine Annual Meeting, 2025, Honolulu, HI, USA*. p0677. **(Summa Cum Laude & AMPC Selected)**
4. **Dai Q**, Chiang J, Nyborg G, Shih SF, Lu DS, Wu HH. Real-time MR thermometry to monitor microwave ablation and predict the ablation zone: Validation in an oncopig cancer model using immediate post-ablation MRI and gross pathology. *International Society for Magnetic Resonance in Medicine Annual Meeting, 2025, Honolulu, HI, USA*. p2189.
5. Zhou W, **Dai Q**, Curiel O, Lu DS, Chiang J, Tsao TC, Wu HH. 3D keypoint-based neural network for rapid needle localization on multislice 2D MRI. *International Society for Magnetic Resonance in Medicine Annual Meeting, 2025, Honolulu, HI, USA*. p2040.
6. **Dai Q**, Nyborg G, Shih SF, Pham J, Lu DS, Chiang J, Wu HH. MR-guided hepatic microwave ablation with real-time MRI guidance and MR thermometry: Initial experience in an oncopig cancer model. *14th Interventional MRI Symposium, 2024, Annapolis, MD, USA*.
7. **Dai Q**, Shih SF, Curiel O, Chiang J, Lu DS, Tsao TC, Wu HH. Dynamic 3D thermometry in moving tissue using accelerated stack-of-radial MRI and an image-navigated multi-baseline PRF method. *International Society for Magnetic Resonance in Medicine Annual Meeting, 2024, Singapore*. p1277. **(Summa Cum Laude & AMPC Selected)**
8. **Dai Q**, Shih SF, Zhou J, Zhang L, Wu HH. Dynamic 3D stack-of-radial multi-baseline PRF MR thermometry using compressed sensing reconstruction and image-based navigation. *International Society for Magnetic Resonance in Medicine Annual Meeting, 2023, Toronto, ON, Canada*. p4987. **(Summa Cum Laude)**
9. **Dai Q**, Zhang L, Li X, Yu P, Tsao TC, Wu HH. Dynamic 3D stack-of-radial PRF MR thermometry to monitor high-intensity focused ultrasound heating: Validation in a tissue motion phantom. *International Society for Magnetic Resonance in Medicine Annual Meeting, 2022, London, UK*. p4084.

10. **Dai Q**, Tang S, Meng MV, Slater JB, Gordon JW, Vigneron DB, Larson PEZ, Wang ZJ. Initial experience of hyperpolarized ^{13}C pyruvate MRI in patients with renal tumors. *Hyperpolarized Carbon-13 MRI Technology Development Workshop, 2020, San Francisco, CA, USA*.
11. **Dai Q**, Kramer SP, Leynes AP, Magudia K, Wang ZJ, Larson PEZ. Deep learning-based prediction of renal cancer grading from contrast-enhanced CT. *UCSF Annual Imaging Research Symposium, 2019, San Francisco, CA, USA*.

Selected Talks & Presentations

Accelerate Medical Imaging Research with AI Tools: From Ideas to Publication. Joint Journal Club (MRRL × CVIB), Department of Radiological Science, University of California, Los Angeles, CA, 2026.

Volumetric Thermometry in Moving Tissues using Stack-of-Radial MRI. International Youth Scholars Forum, Ruijin Hospital, Shanghai, China (Online), 2026.